

A 1.3.6-7 Control Systems (HL Only)



A 1.3.6-1.2.7 Control Systems

A control system measures what is happening, compares it against a target, and acts to reduce the difference. That single loop, repeated continuously, sits behind a thermostat holding a room at 20°C and behind a self-driving car keeping its lane. The two learning statements here are paired: the first asks you to describe the components that make such a system work, the second asks you to explain how those components behave in real applications. This page works through the five components and the input, process, output and feedback mechanism, distinguishes open-loop from closed-loop control, and then explains how the same pattern reappears across the applications the syllabus lists.

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Key Questions

- What are the five components of a control system, and what does each one do?
- What do the terms input, process, output and feedback mean?
- What is the difference between an open-loop and a closed-loop system, and why does it matter?
- Why are closed-loop systems used where accuracy and safety are important?

Scope note

The syllabus says the example applications "may include" the eight listed. You are not expected to memorise that list. You should be able to take any everyday automated system and identify its components and whether it relies on feedback.

IB Command Term note

This page covers two command terms.

A1.3.6 asks you to **describe**: give a detailed account of the components and the input, process, output and feedback mechanism. A description names each part and states its function.

A1.3.7 asks you to **explain**: give reasons. For an application, that means saying how the components work together and why feedback produces the behaviour the system needs. An answer that only lists components answers "describe" weakly and does not answer "explain" at all.

What are the components of a control system?

A control system is built from five kinds of component. Each has one job, and they pass information and action around the loop in a fixed order.

The controller

The controller is the decision maker. It takes the target value and the current measurement, works out the difference between them, and decides what to do about it. In a self-driving car the controller is the onboard computer; in a thermostat it is a small circuit comparing the set temperature against the room temperature. Everything else in the system either feeds the controller information or carries out its instructions.

Sensors

A sensor measures some quantity in the system or its environment: temperature, speed, distance, soil moisture, the presence of a person. Without sensors the controller can only follow a fixed routine without any input. The quality of a closed-loop system (see below) depends heavily on its sensors, because a controller can only act on what it can measure.

Transducers

A transducer converts a physical quantity into an electrical signal the controller can read, or converts an electrical signal back into a physical action. A thermistor turning temperature into a voltage is a transducer; so is a loudspeaker turning a signal into sound.

Sensors and transducers overlap. A sensor is the component whose job is to measure, while transducer converts one form of energy into another. Most sensors contain a transducer to do their measuring.

Actuators

An actuator carries out the controller's decision by acting on the physical world. It is the opposite end of the loop from the sensor. The sensor measures, the actuator changes things. Motors, valves,

heating elements, and the throttle and brakes of a car are all actuators. A controller's decision has no effect until an actuator turns it into movement, heat or flow.

The control algorithm

The control algorithm is the set of rules the controller follows to turn measurements into decisions. At its simplest it is a rule like "if the temperature is below the target, switch the heater on."

More capable systems weigh how large the error is and how fast it is changing, so they can act smoothly rather than snapping fully on and off.

The algorithm is where the system's intelligence lives: the same sensors and actuators behave very differently under a better algorithm.

COMMON EXAM MISTAKE

Treating "sensor" and "transducer" as the same thing. A sensor is defined by its job, which is to measure. A transducer is defined by what it does to a signal, which is to convert one form of energy into another. Most sensors use a transducer to measure, but a question asking for the role of a transducer wants the idea of conversion, not measurement.

Input, process, output and feedback

Every control system can be described with four terms.

The **input**, or setpoint, is the target you want: 20°C, 65 mph, a closed door.

The **process** is the thing being controlled, such as the room, the car or the door.

The **output** is the actual value the process currently has, which may or may not match the input.

Feedback is the measured output sent back to the controller so it can compare actual against target. The comparison of input against feedback produces the error, and reducing that error is the system's whole job.

THINK FIRST

A toaster on a timer and a central heating thermostat both control something. The toaster runs for a set time whatever the bread is doing; the thermostat keeps measuring the room and reacts. What can the thermostat do that the toaster cannot, and why might that matter on a cold day?

WORKED EXAMPLE

A home thermostat with the target set to 20°C.

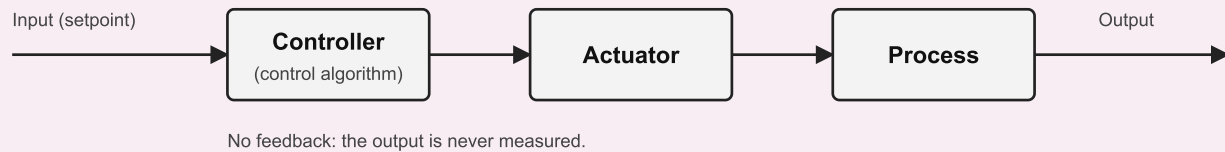
1. **Input:** the setpoint is 20°C.
2. The **sensor** measures the room at 18°C, and this is fed back to the controller.
3. The **controller** compares: the room is 2°C below target, so there is a positive error.
4. The control **algorithm** decides the heating should be on.
5. The **actuator** switches the boiler on, and the room warms. This is the process.
6. The **sensor** now reads 20°C, the error reaches zero, and the controller switches the heating off.
7. The room cools slightly, the error grows again, and the **loop repeats**.

The system never sets the temperature once and forgets it. It holds the target by measuring and correcting continuously, which is exactly what the timer-based toaster cannot do.

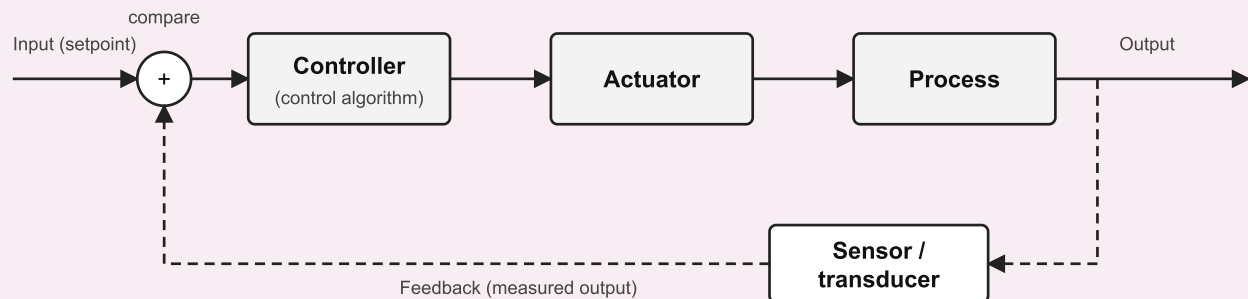
Open-loop and closed-loop control

The most important distinction in this topic is whether the system uses feedback. An open-loop system sends its commands and never checks the result. A closed-loop system measures the output and feeds it back, so it can correct itself. A toaster is open-loop. A thermostat is closed-loop. This is the distinction an examiner can build a question around most easily, so be sure you can state it and give an example of each.

Open-loop control



Closed-loop control



Open-loop versus closed-loop control. The feedback path, present only in the closed-loop system, is what allows it to measure its output and correct itself.

Feature	Open-loop	Closed-loop
Feedback	None; the output is not measured	Output measured and fed back
Accuracy	Lower; cannot correct errors	Higher; corrects itself
Adapts to disturbance	No	Yes
Complexity and cost	Simpler, cheaper	More complex, more expensive
Example	Timer sprinkler, basic toaster	Thermostat, cruise control

Where a mistake is costly, the ability to detect and correct error is worth the extra complexity. That is why braking, flight control and medical infusion pumps are closed-loop, while a system whose job is trivial or whose conditions never change can stay open-loop and cheaper.

How control systems work in real applications

The same loop appears across very different machines. The skill the syllabus wants is to look at an application and explain it in these terms: what is sensed, what decides, what acts, and whether feedback closes the loop.

A **self-driving car** senses the road with cameras, lidar and radar; transducers turn those readings into data; the controller, running navigation and obstacle-avoidance algorithms, decides on a speed and heading; actuators apply the steering, throttle and brakes. Feedback is constant, because the next frame of camera data shows the result of the last decision. This is closed-loop control under hard real-time constraints, which is why sensor failure is so serious: a blind controller cannot correct.

An **irrigation controller** senses soil moisture and, in better systems, rainfall. The control algorithm waters only when moisture drops below a threshold, so a wet week produces no watering and a dry one produces more. The actuator is a valve. The feedback, soil moisture, is what separates this from a simple timer that waters on a fixed schedule regardless of the weather.

Traffic signals sense waiting vehicles with inductive loops or cameras and adjust their timings to clear the busiest approach. Traffic flow feeds back through the same detectors, so the system responds to demand rather than running a fixed cycle.

HOW IT WORKS (CONCEPTUALLY)

These are the eight applications the syllabus lists, each as the same loop.

Application	Senses	Acts with	Uses feedback?
Autonomous vehicle	Cameras, lidar, radar	Steering, throttle, brakes	Yes
Home thermostat	Temperature	Boiler or cooler	Yes
Automatic lift	Position, weight	Motor, doors	Yes
Washing machine	Water level, temperature, load	Valves, heater, drum motor	Yes
Traffic signals	Vehicle detectors, cameras	Signal lamps	Yes
Irrigation	Soil moisture, rainfall	Water valves	Yes
Home security	Motion, door and window sensors	Alarm, lights, cameras	Yes; detect then respond
Automatic doors	Motion or pressure	Door motor	Opening is closed-loop; closing is

Application	Senses	Acts with	Uses feedback?
			often a fixed timer (open-loop)

Summary

Use this summary to revise and remember the key points.

Term	What it is
Controller	Compares target with measurement and decides what to do
Sensor	Measures a quantity in the system or environment
Transducer	Converts a physical quantity to or from an electrical signal
Actuator	Carries out the controller's decision in the physical world
Control algorithm	The rules turning measurements into decisions
Input (setpoint)	The target value
Process	The thing being controlled
Output	The actual current value
Feedback	Measured output returned to the controller
Open-loop	No feedback; cannot self-correct
Closed-loop	Uses feedback; corrects itself

TRY IT ON PAPER

The quiz below is multiple choice, so use these exam style questions to practise writing the command terms. Write full answers on paper (or a laptop if you use one in the exam), then check against the model answers.

1. Describe the role of each of the following in a control system: sensor, controller, actuator. [3 marks]

Model answer: The sensor measures a quantity in the system or its environment [1]. The controller compares the measured value with the target and decides what action to take [1]. The actuator carries out that decision by acting on the physical world, for example a motor or valve [1].

2. A greenhouse uses a closed-loop system to hold its temperature at 24°C. Explain how the components work together to maintain this temperature, and why a closed-loop design is used rather than open-loop. [4 marks]

Model answer: A temperature sensor measures the greenhouse air and feeds this back to the controller [1]. The controller compares the measurement with the 24°C setpoint and, using its control algorithm, decides whether to heat, ventilate or do nothing [1]. An actuator such as a heater or vent motor carries out the decision, changing the temperature, which the sensor then measures again [1]. A closed-loop design is used because feedback lets the system correct for disturbances such as a cold night or strong sun, which an open-loop timer could not respond to [1].

Quiz Control Systems

1

Which component of a control system is responsible for gathering information about the environment?

A. ☒ Sensor

B. ☐ Control algorithm

C. ☐ Actuator

D. ☐ Controller

The sensor's job is to measure or detect a quantity in the environment. The control algorithm processes that information but does not gather it, the actuator acts on the world, and the controller makes decisions.

2

What is the role of a transducer in a control system?

- A. ☒ To convert energy
- B. ☐ To make decisions
- C. ☐ To gather information
- D. ☐ To perform actions

A transducer converts one form of energy into another, typically a physical quantity such as temperature or pressure into an electrical signal the controller can read. Gathering information describes a sensor's purpose; the two overlap, which is why this is tested.

3

Which type of control system uses feedback to adjust its actions?

- A. ☐ Straight-feed
- B. ☐ Feed-loop
- C. ☒ Closed-loop
- D. ☐ Open-loop

A closed-loop system feeds the measured output back to the controller so it can adjust. An open-loop system sends commands without checking the result.

4

Which of the following is NOT an example of a control system?

- A. ☐ A traffic light
- B. ☐ An automatic washing machine

C. ☐ A home thermostat

D. ☒ A manual door

A manual door has no controller, sensors or feedback; a person operates it directly. The other three sense conditions and act on them automatically.

5

What is the "input" in a control system?

A. ☐ The action taken by the system

B. ☐ The feedback from the system

C. ☒ The desired state of the system

D. ☐ The current state of the system

The input, or setpoint, is the target the system is trying to achieve. The current state is the output, the action is what the actuator does, and feedback is the measured output returned to the controller.

6

Which of the following is an advantage of a closed-loop system?

A. ☒ Accuracy

B. ☐ No need for sensors

C. ☐ Low cost

D. ☐ Simplicity

Feedback lets a closed-loop system detect and correct error, so it is more accurate. Low cost and simplicity are advantages of open-loop systems, and a closed-loop system cannot work

without sensors.

7

What is the role of the controller in a control system?

- A. ☐ To convert energy
- B. ☒ To make decisions
- C. ☐ To perform actions
- D. ☐ To gather information

The controller compares the measurement with the target and decides what to do. Converting energy is a transducer's role, performing actions is an actuator's, and gathering information is a sensor's.

8

Which of the following is an example of an actuator in a car's cruise control system?

- A. ☐ Steering wheel
- B. ☒ Throttle
- C. ☐ GPS
- D. ☐ Speed sensor

The throttle acts on the car to change its speed, which is what an actuator does. GPS and the speed sensor gather information, and the steering wheel is not part of speed control.

9

Which control system is more likely to be used in a safety-critical application?

- A. ☐ Both are equally suitable

B. ☐ Neither is suitable

C. ☒ Closed-loop

D. ☐ Open-loop

A closed-loop system can detect and correct error through feedback, which matters when failure is dangerous. An open-loop system cannot respond to anything going wrong.

10

Which of the following is NOT a real-world application of control systems?

A. ☐ Autonomous vehicles

B. ☐ Automatic doors

C. ☐ Home security systems

D. ☒ Manual can opener

A manual can opener is operated entirely by hand, with no sensing, decision or feedback. The other three are automated control systems.

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